Abhijeet Malviya

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Professional Summary

AI/ML and software engineer with 3+ years of experience building production AI systems, distributed services, and secure data platforms. Experienced in LLMs, RAG, machine unlearning, adversarial ML, Java, Python, Spring Boot, FastAPI, PostgreSQL, AWS, Docker, and Kubernetes. M.S. thesis researcher focused on cybersecurity and machine unlearning. Focused on building reliable, scalable AI systems that bridge applied machine learning, backend engineering, and security.

Education

University of Nevada, Las Vegas — Las Vegas, NV
Master of Science in Computer Science – Thesis Track, Cybersecurity & Cryptography — *Anticipated May 2027*
- Advisor: Dr. Son. Coursework: Private & Secure AI, AI-Powered Cyber Attacks & Defenses, Machine Learning, Compilers
- Teaching Assistant / Grader for CS 115 and CS 138

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Engineering — *August 2019 – May 2022*

Research & Publications

- **M.S. Thesis**, Cybersecurity & Cryptography, UNLV – *Machine Unlearning for Cybersecurity Applications*; defense Spring 2027
- Researching machine unlearning methods for selectively removing training-data influence while preserving model utility, with applications to cybersecurity, privacy, and secure AI
- **Book Chapter Contributor** – *"Automation Is Not Autonomy,"* Chapter 5 in *Security Singularities: Building Adaptive, Autonomous Cyber Defense Systems for the Modern Digital Enterprise*

Experience

UNLV — August 2025 – Present
AI/ML Research Engineer | Alzheimer's Disease Biomarker Observatory — *Las Vegas, NV*
- Architected role-based access control and multi-tenant isolation for a platform handling sensitive patient data across institutions, enforcing per-institution boundaries and auditable access paths
- Engineered RAG retrieval over 10K+ biomedical documents (pgvector, transformer embeddings) at <2s latency; added retrieval allow-listing so cross-institution documents could not leak via semantic search
- Deployed agentic research platform (Next.js, Python, PostgreSQL) enabling automated anomaly detection and root-cause analysis for 50+ researchers

Resorts World Las Vegas — November 2024 – July 2025
AI/ML Engineer — *Las Vegas, NV*
- Developed and deployed 100+ LLM-powered automation services (Groovy, SQL, LLMs), improving intent-classification performance and reducing p95 inference latency by 40%
- Built scalable ML APIs and automation pipelines for Opera Cloud, increasing data integration throughput 40% and resolving 5K+ weekly customer queries

Denken Solutions — March 2023 – November 2024
Software Developer — *Irvine, CA*
- Built distributed Apache Spark pipeline for adversarial entity detection, surfacing 2,200+ malicious entities daily – trust-and-safety detection at production scale
- Designed event-driven microservices (Java 21, Spring Boot, Kafka) sustaining 2K+ TPS at 180ms p99 latency
- Automated CI/CD with Docker, Kubernetes, Terraform, and AWS EKS enabling zero-downtime deployment

Incedo — September 2022 – March 2023
Software Engineer — *Florham Park, NJ*
- Developed and optimized RESTful APIs and full-stack features serving 1M+ users using Java, JavaScript, and SQL
- Implemented observability stack (OpenTelemetry, Prometheus, Grafana) reducing MTTR 60% via real-time anomaly detection and alerting

Projects

RAG-as-a-Service - *Java 21, Spring Boot, LangChain4j, pgvector, Docker*
- Built Java-based RAG platform with Tika ingestion, recursive chunking, pgvector embeddings, similarity-filtered retrieval, and source citations
- Implemented session-scoped conversation memory and similarity thresholds to reduce ungrounded LLM responses
- Containerized the retrieval service with Docker for reproducible deployment and downstream application integration

PreMarket AI - *FastAPI, PostgreSQL, WebSockets, LLMs*
- Built multi-agent LLM system for real-time pre-market stock scoring, anomaly detection, and forecasting
- Architected multi-user FastAPI backend with PostgreSQL pipelines for ingesting, storing, and scoring high-frequency market data
- Implemented WebSocket streaming for low-latency delivery of live market signals and LLM-generated analysis